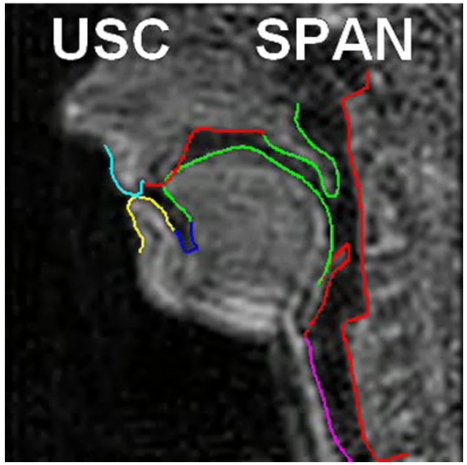
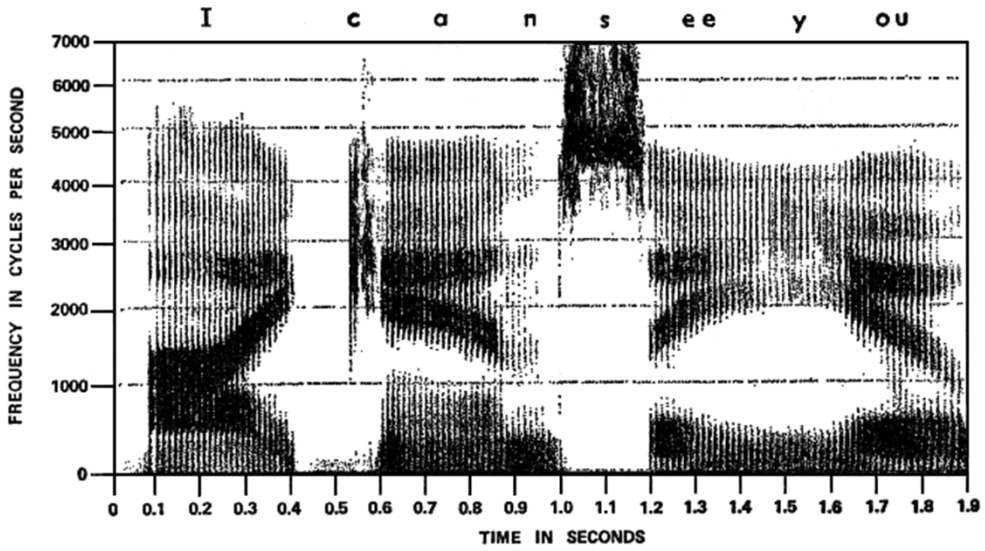
Speech Perception I: Categorical Perception

What is speech perception?

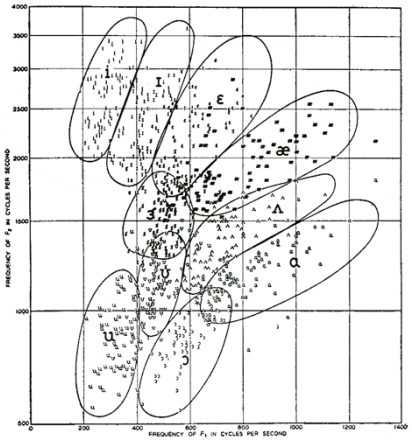
Somehow we are able to take this mess and retrieve a message:





We also extract *the same message* from very different physical stimuli:

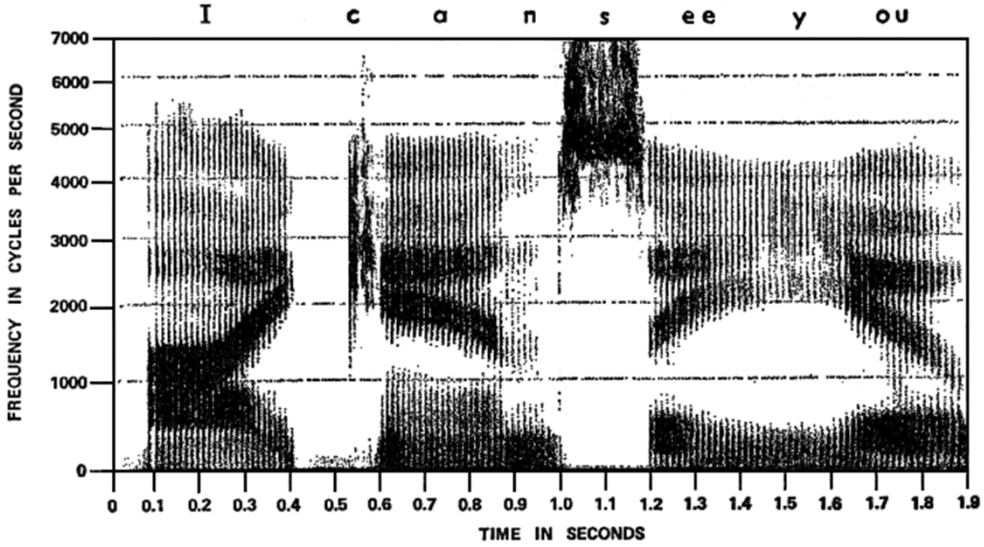




Frequency of second formant *versus* frequency of first formant for ten vowels by 76 speakers.

One premise:

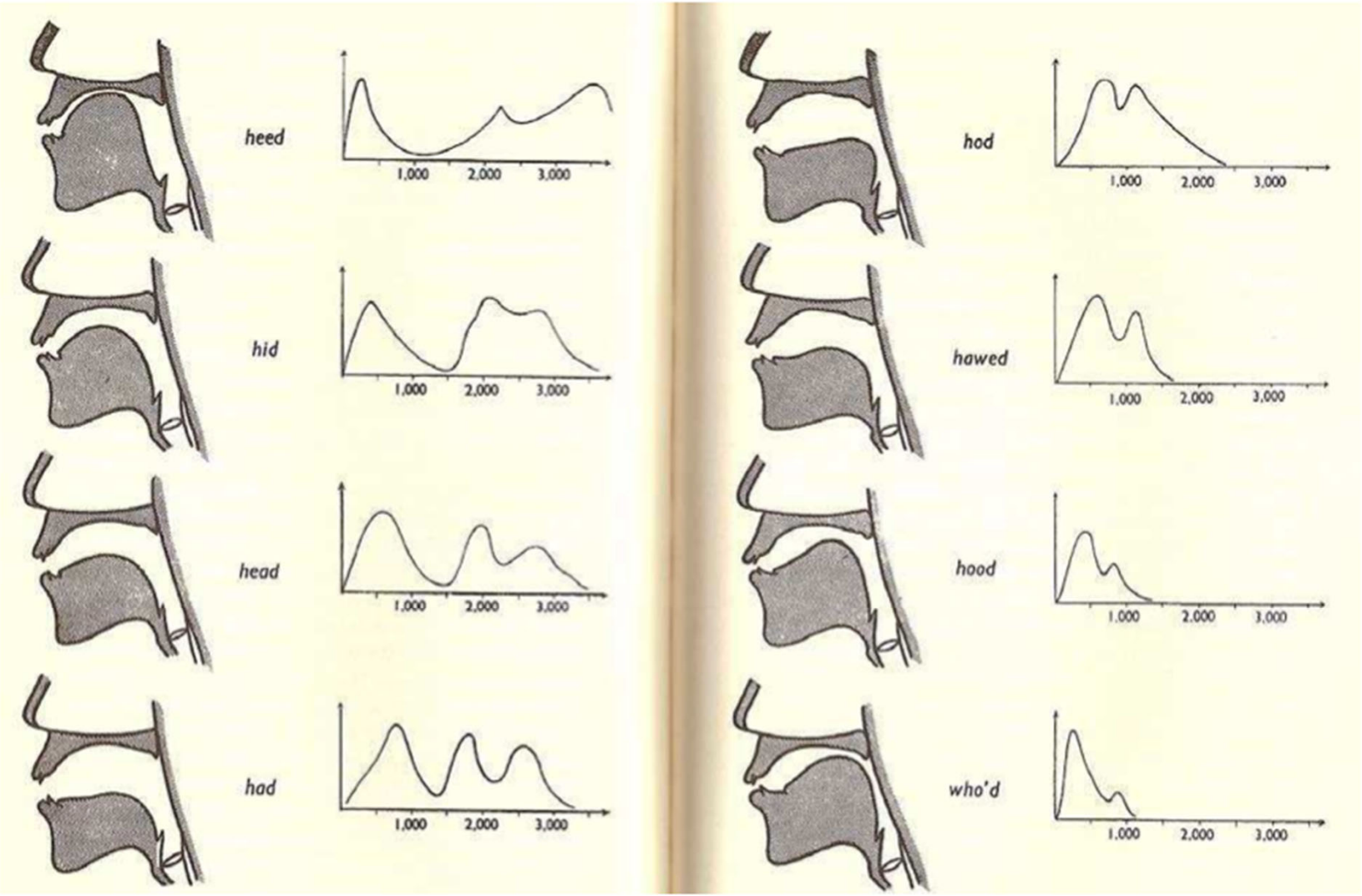
Somehow we have to turn a highly variable, analog signal, into something more abstract, so that the language system can operate on it.



Categorical perception is thought to reflect this kind of "translation" from physical to symbolic.

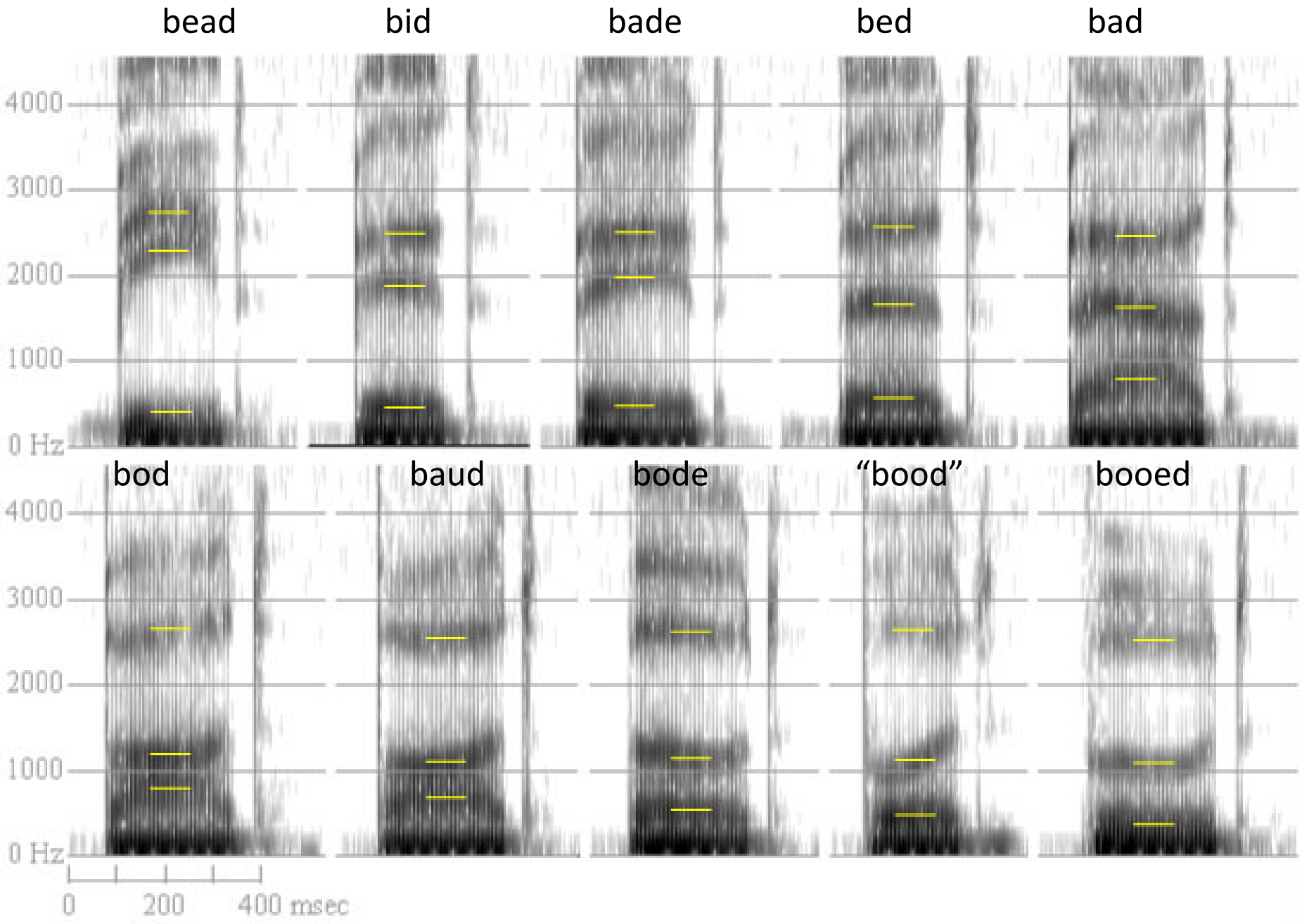
Note, next time, we'll talk about some alternative ideas about this.

Formants and formant transitions



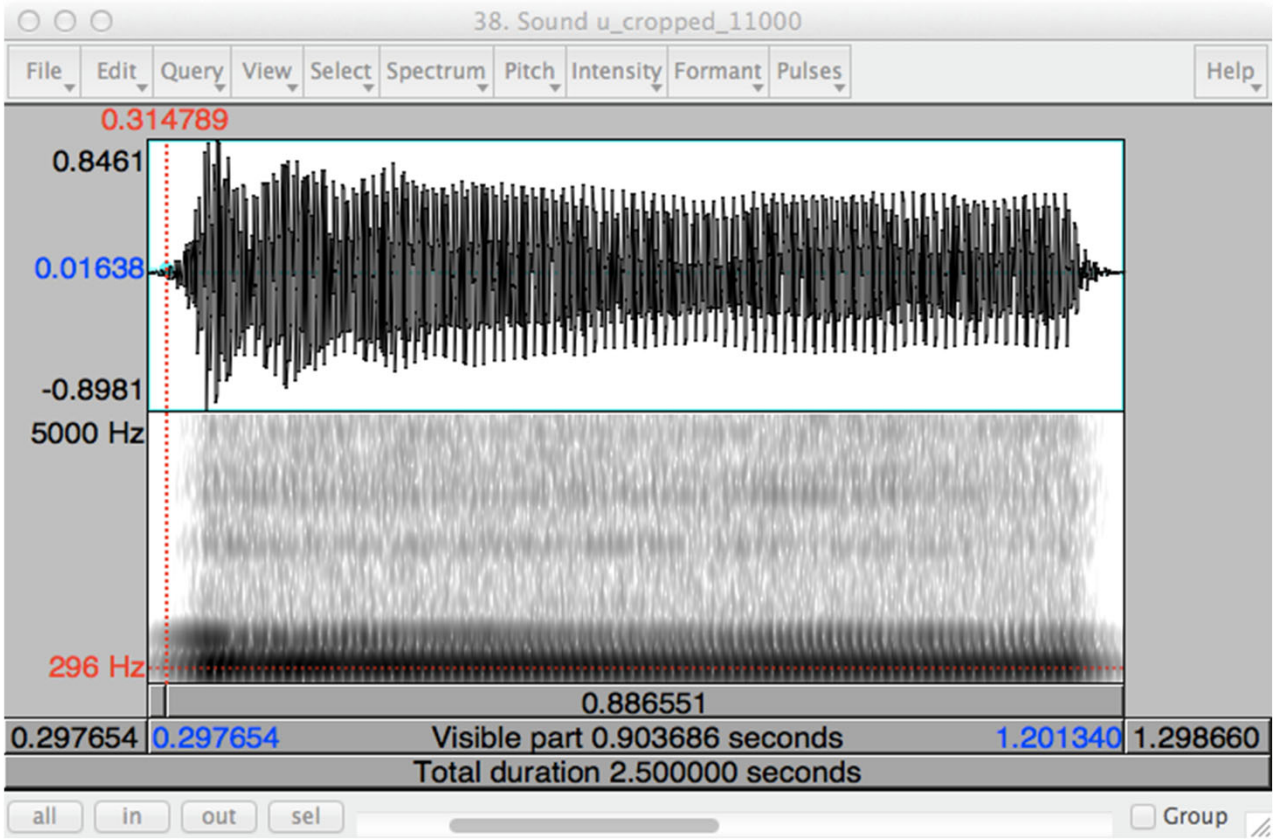
The shape of the vocal tract is related in lawful ways to the amplification of particular resonant frequencies (from Ladefoged, 1996).

Here they are as spectrograms



English vowels from Rob Hagiwara: (http://home.cc.umanitoba.ca/~robh/howto.html#formants)

Let's learn about formants by doing some source-filter synthesis



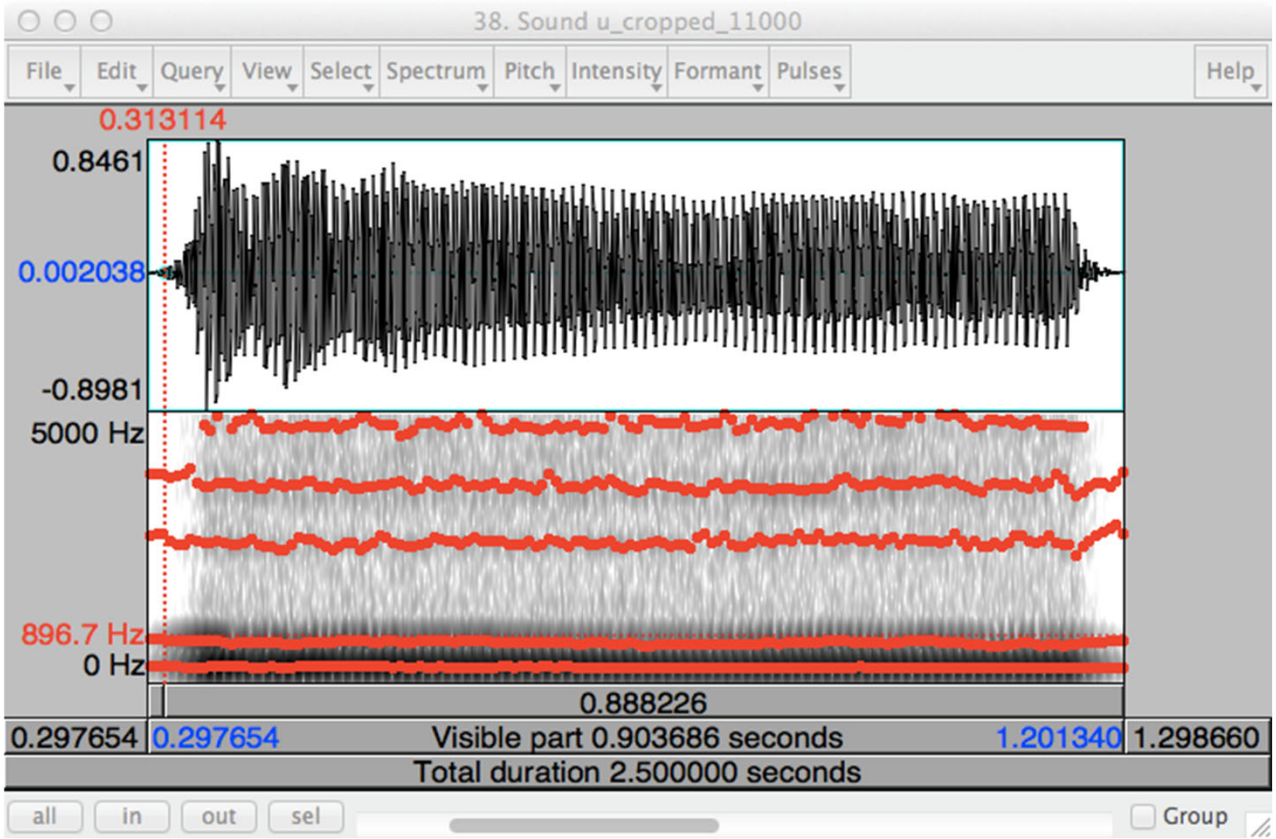
This is a spectrogram of me saying /u/.

I've got the cursor on the first formant, you can see it's at about 300Hz.

The second formant is very low, around 900Hz.

(I did all of this with Praat free software for speech analysis, available at praat.org)

Let’s learn about formants by doing some source-filter synthesis



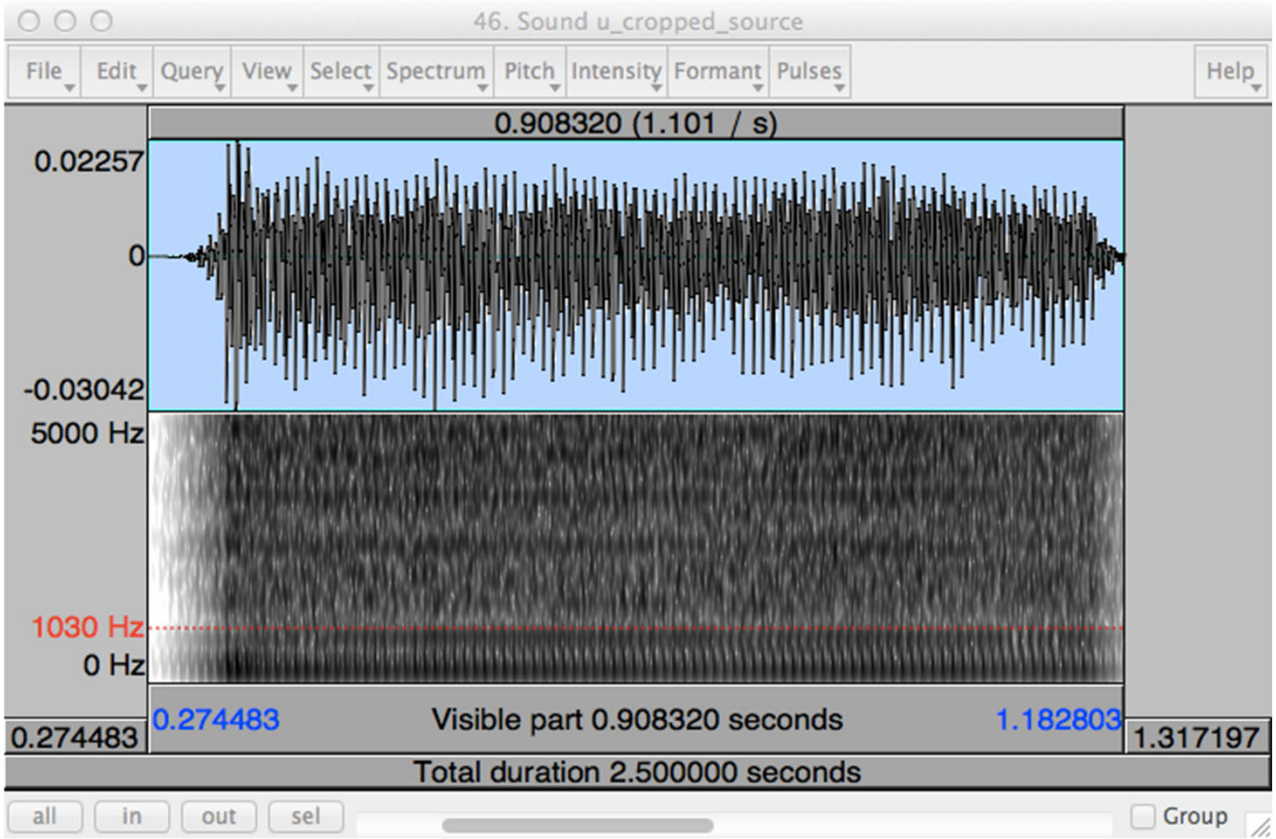
Actually, Praat can automatically find formants.

Here it’s identified five.

Only the first three are usually important for phonemic processing.

In this example, we’ll only deal with the first two.

This is the source…

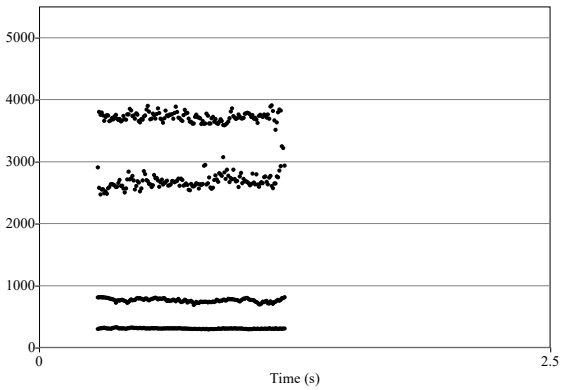


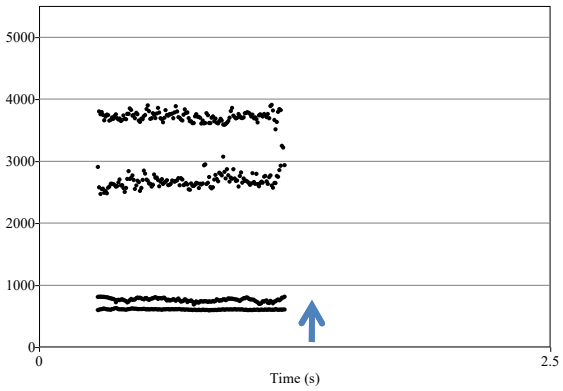
So, now what I've done is extract the SOURCE from the sound, using a model of the filter based on the formant frequencies.

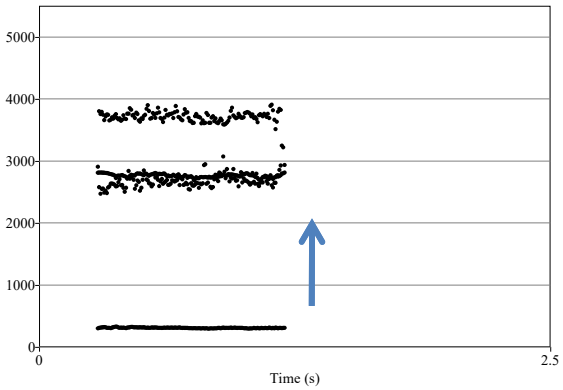
(This is a little bit like doing a multiple regression analysis.)

The result is a more-or-less "unfiltered" version of my voice. It sounds like a buzz…

…and these are the filters!







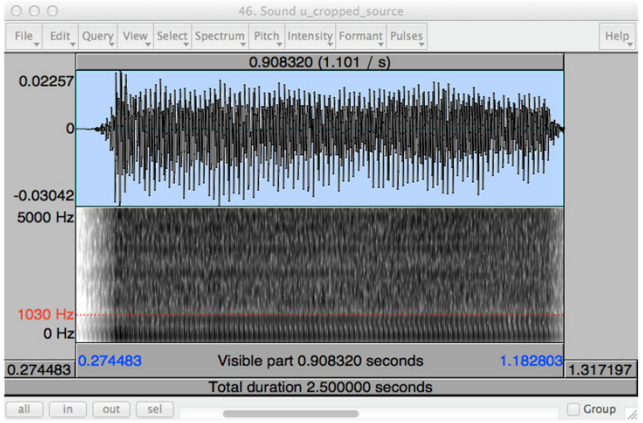
Shown here are three different filters we can apply to our source.

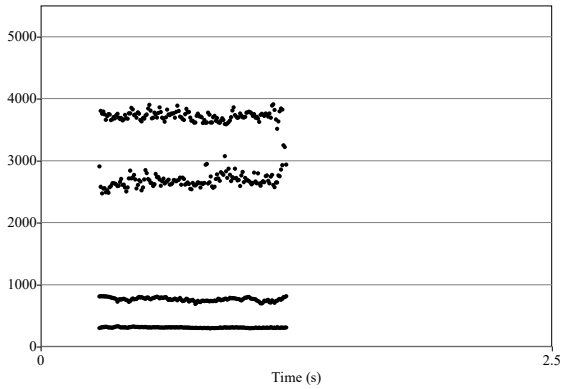
The first is just the first four formants from the original recording.

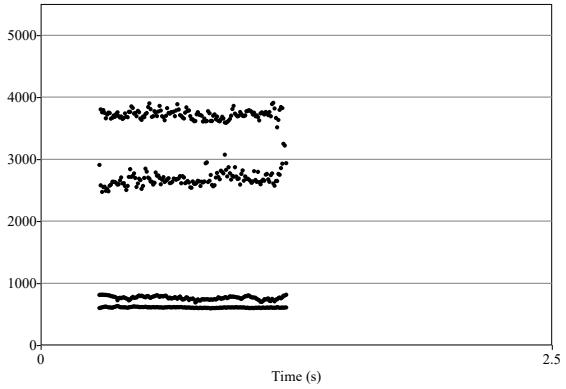
In the second one, I've just raised the first formant a little.

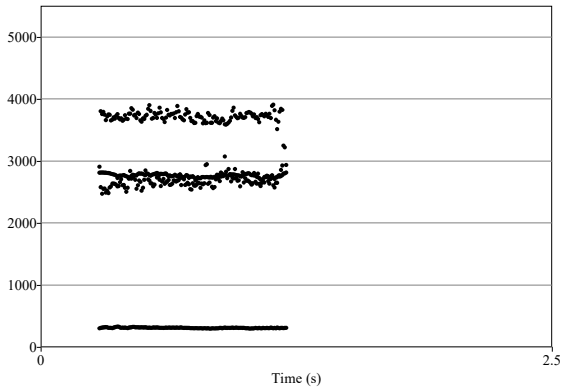
In the second one, I've raised the second formant a lot.

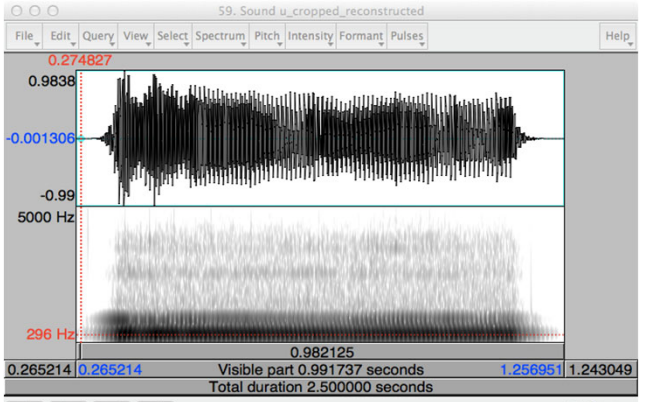
Multiplying the source by the filters gives us new vowels.

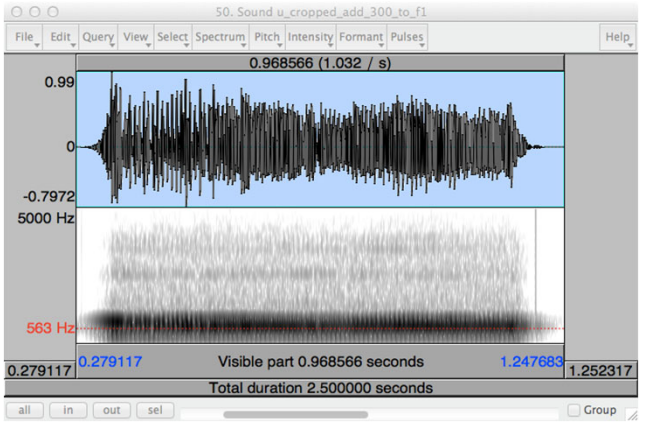


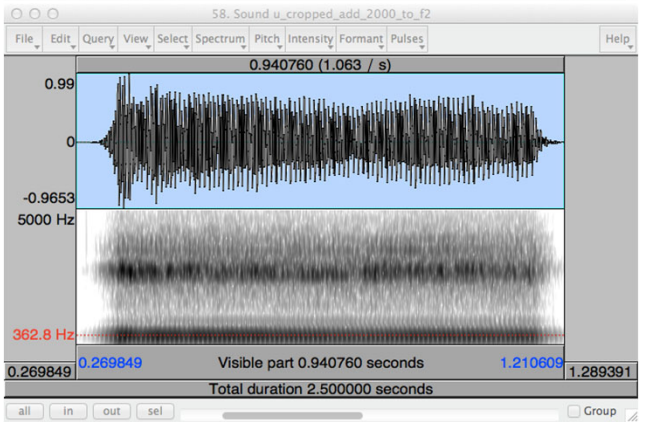








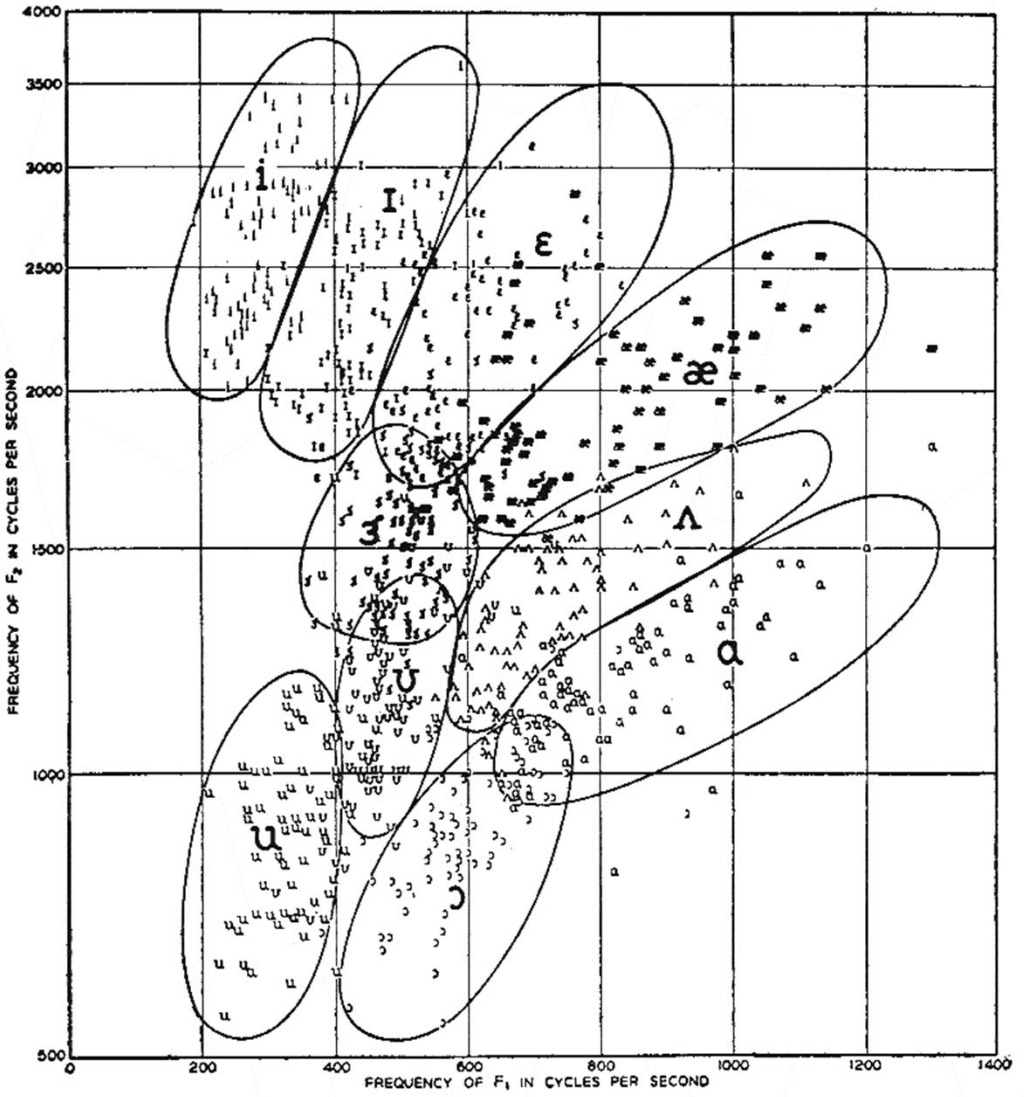




Remember that formant values are partly determined by anatomical features that differ between speakers of the same language, by different rates of speech, and by context (e.g., the phones produced before and after a vowel).

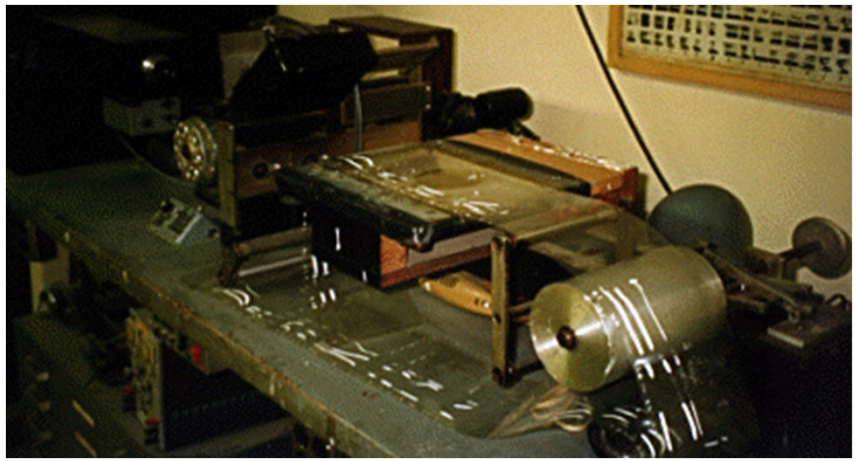
Nonetheless, formant values can provide a pretty good description of “vowel space,” especially for a specific talker, speaking carefully.

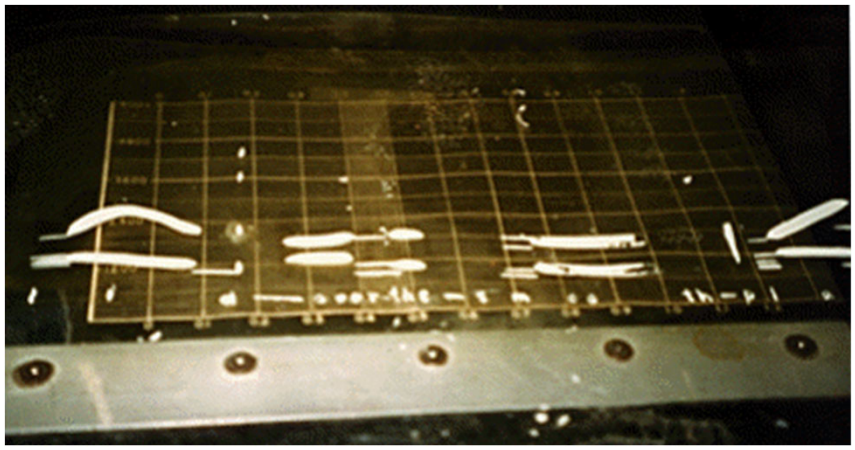
All else being equal, we can produce reasonable approximations of vowels by manipulating steady-state formants.

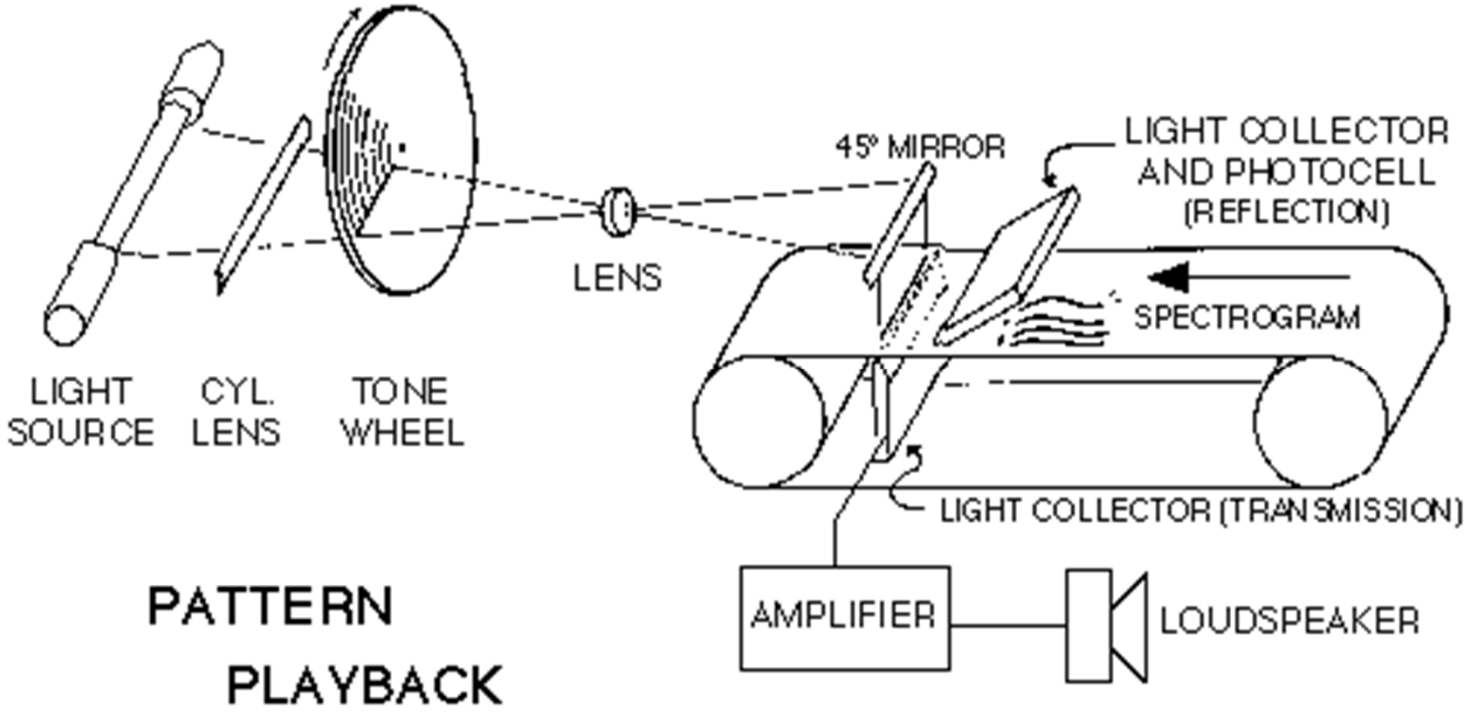


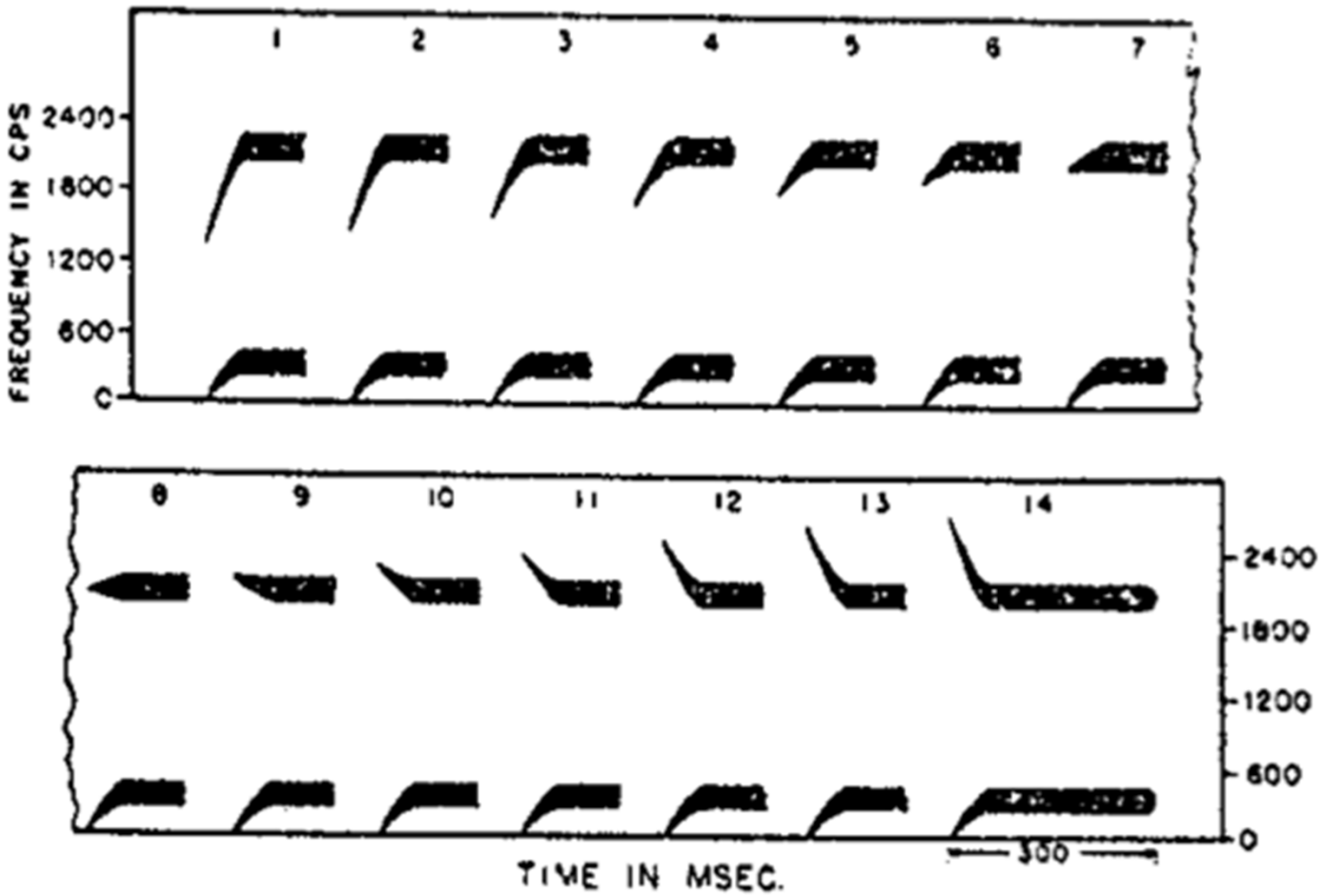
Frequency of second formant *versus* frequency of firs formant for ten vowels by 76 speakers.

The “pattern playback” synthesizer.







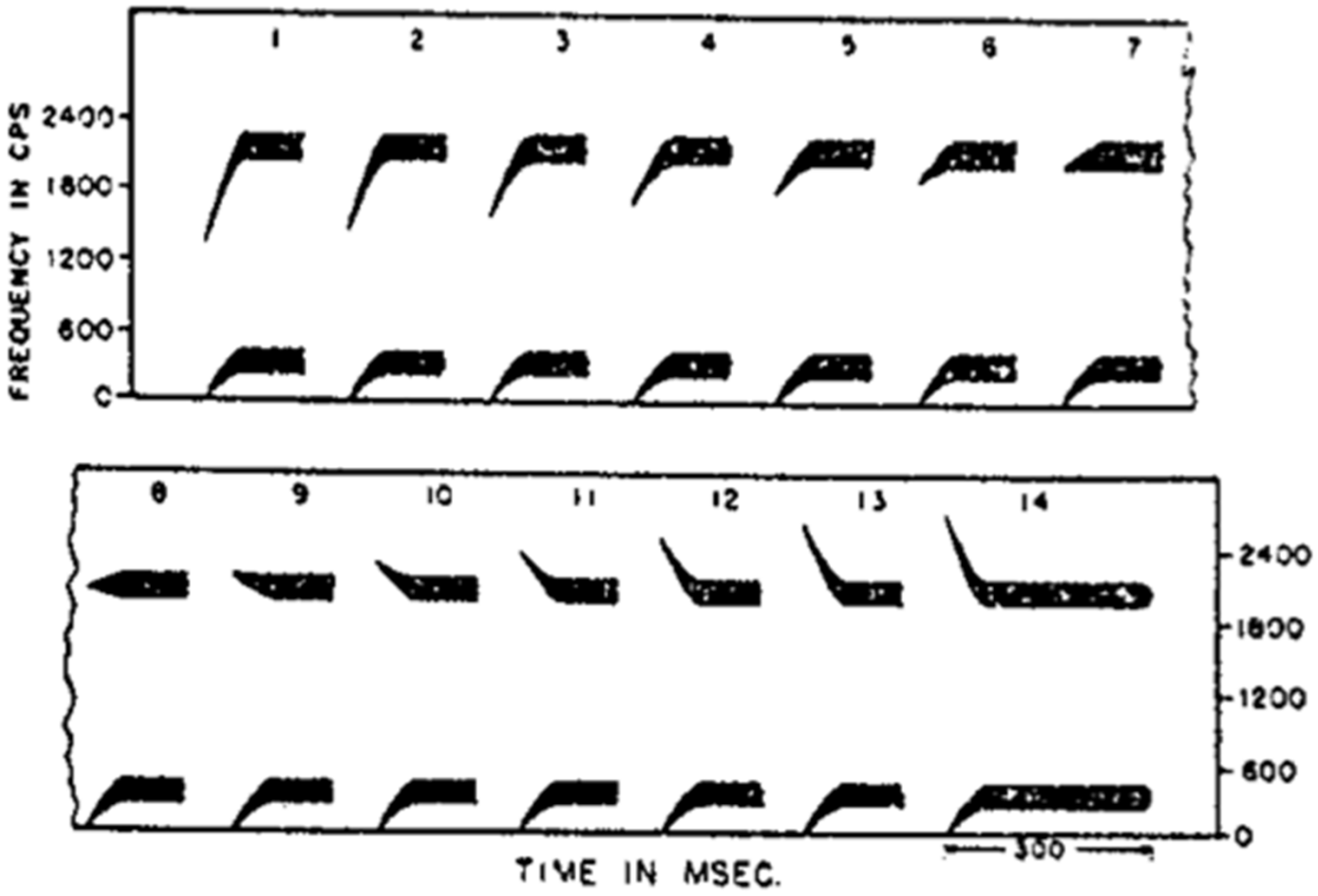


Stop consonants are distinguished by *formant transitions*.

This is sensible if you think about how they're produced: the audible part of a /b/, /d/, or /g/, happens while your jaw is opening.

The pattern playback synthesizer was used to create speech-like sounds that replicated these formant transitions.

Liberman et al. (1957) available here: http://www.haskins.yale.edu/Reprints/HL0022.pdf

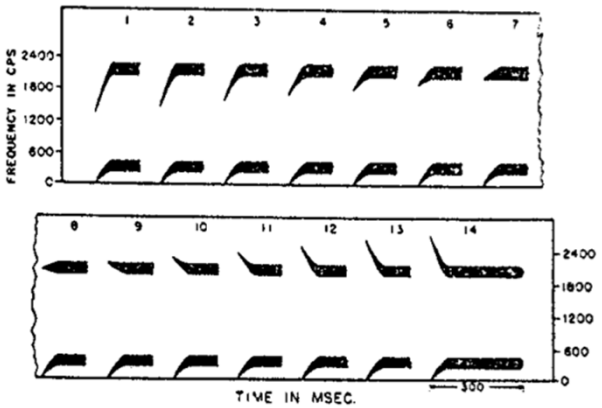


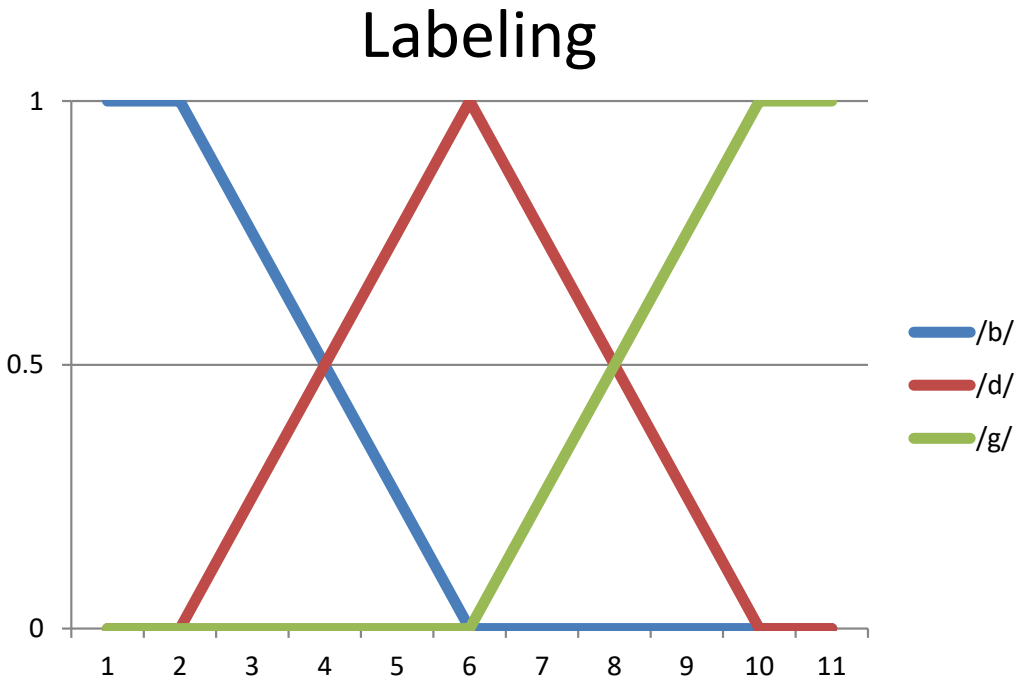
A speech ***continuum****:*

These stimuli are constructed by keeping F1 constant and varying F2 onset frequency *linearly* from stimulus 1 to 13.

Thus: adjacent stimuli differ by *the same amount* across the entire continuum.

A simple “ideal observer” model assuming perception is *non-categorical*.

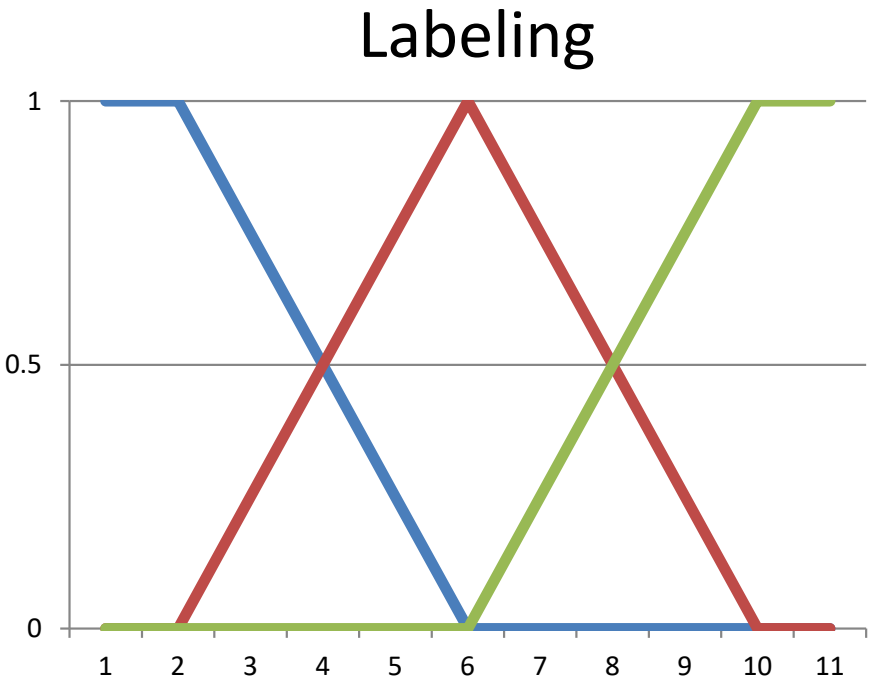


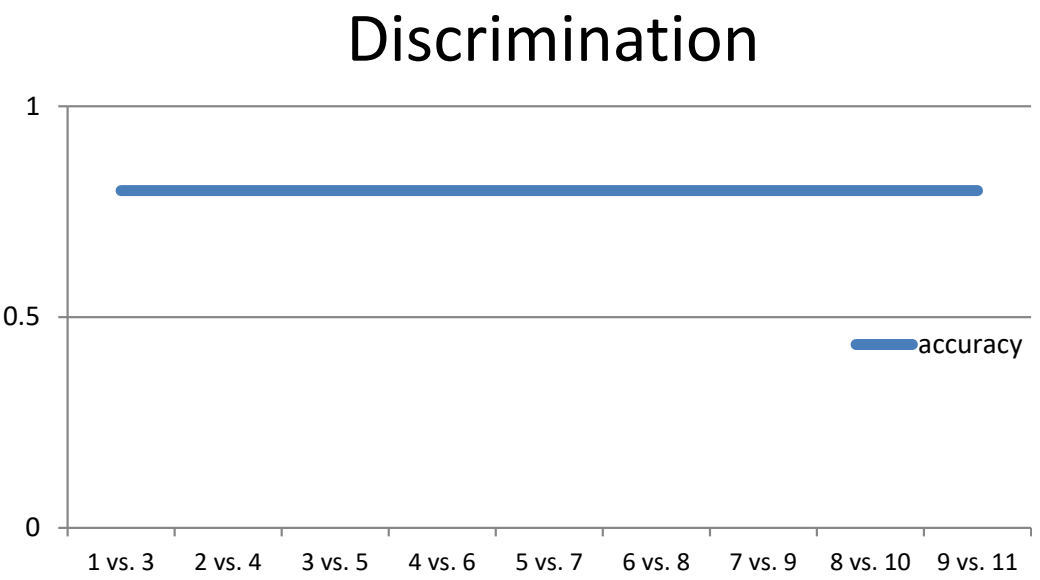


The first thing we’d predict is that categorization would get “blurry” as you move away from category prototypes. We know the endpoints are /b/ and /g/, with /d/ in the middle. Between these points, stimulus ambiguity should scale with the distance from the prototype. So, stimulus 2 would be 75/25 /b/ - /d/.

Liberman et al. (1957) available here: http://www.haskins.yale.edu/Reprints/HL0022.pdf

A simple “ideal observer” model assuming perception is linear.

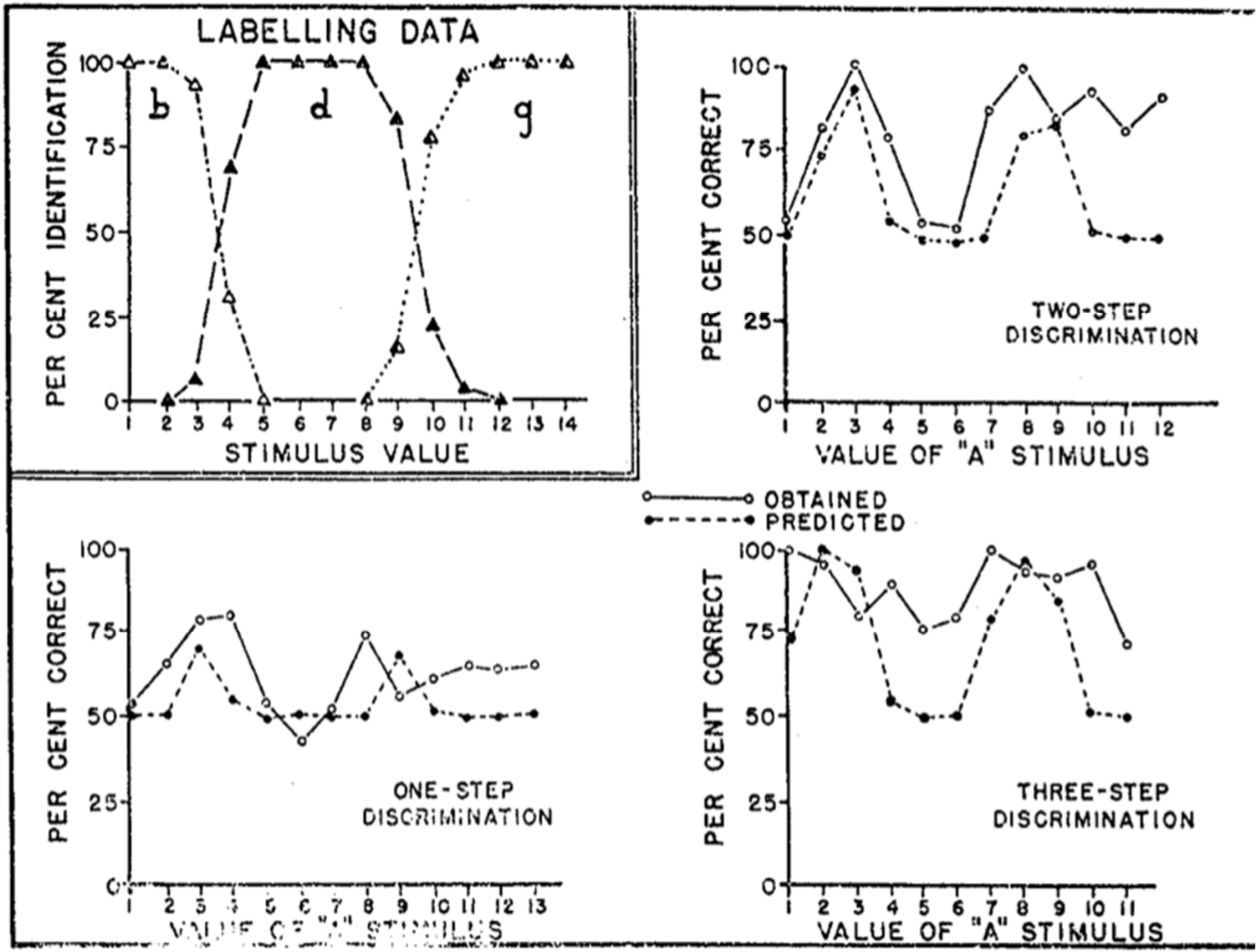




But the other important thing we’d predict is that Discrimination performance should be independent of Labeling.

Obviously labeling has to have a sort of “peaked” function, because you’ve got three categories, and you’re asking people to make binary responses. But for the discrimination task, you’re just asking people to say “same” or “different.” If perception of differences in F2 onset were constant over the whole continuum, discrimination accuracy would also be constant.

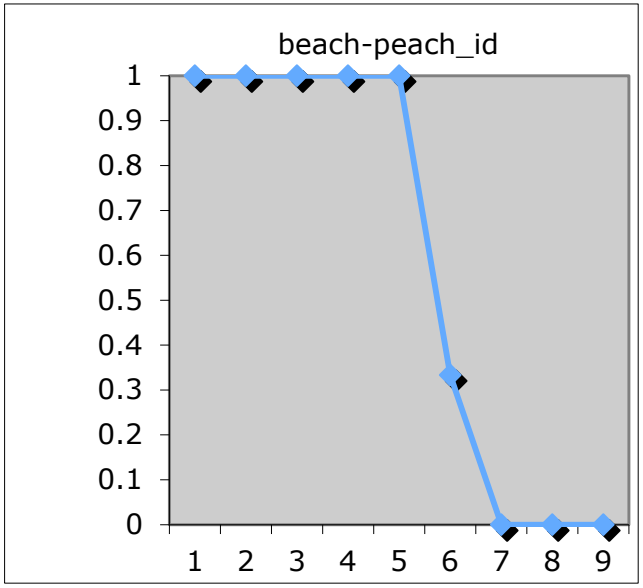
Categorical perception

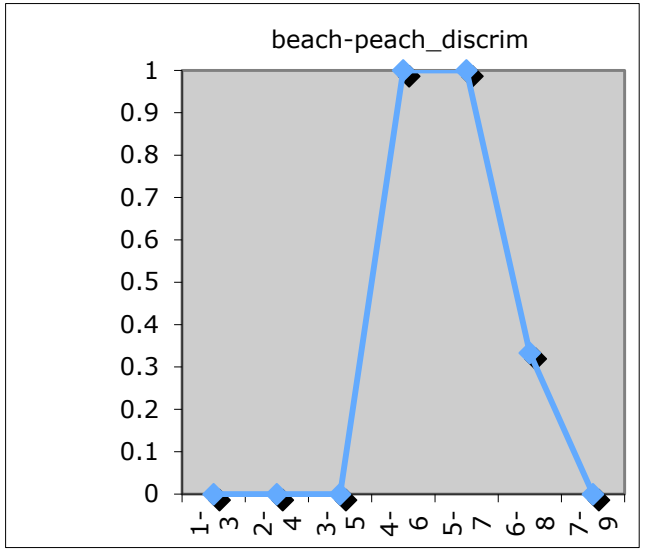


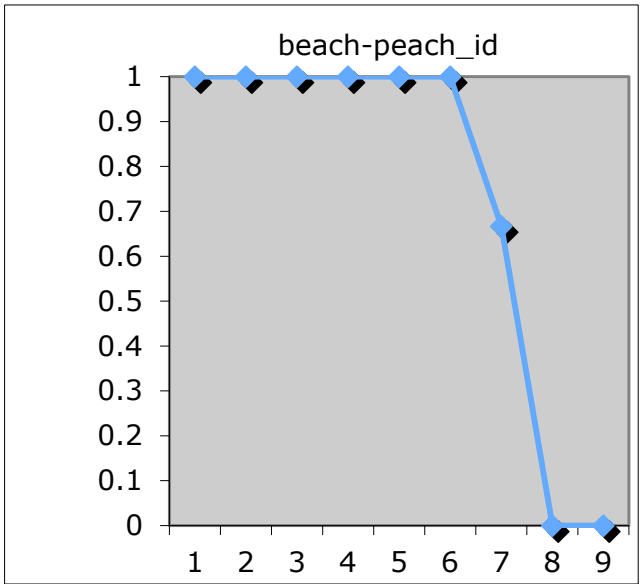
Liberman et al. (1957) available here: http://www.haskins.yale.edu/Reprints/HL0022.pdf

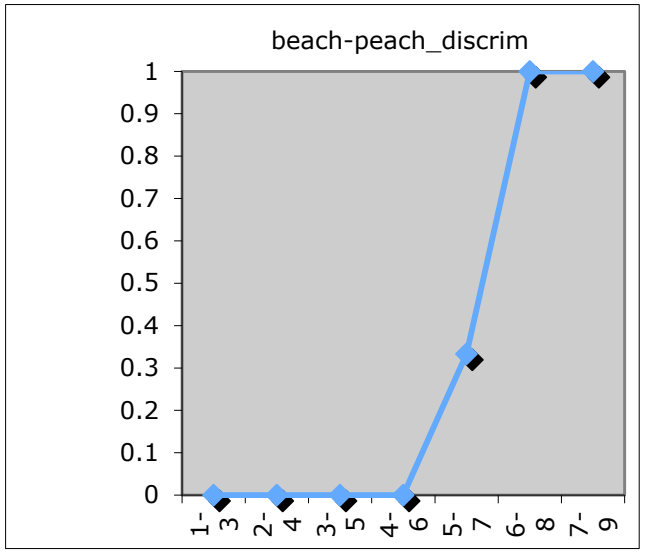
Categorical perception demonstration

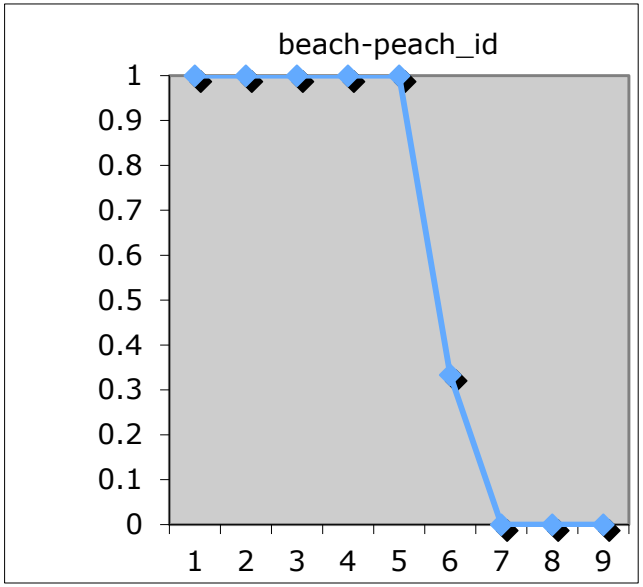
Categorical perception: Class data (prior years)

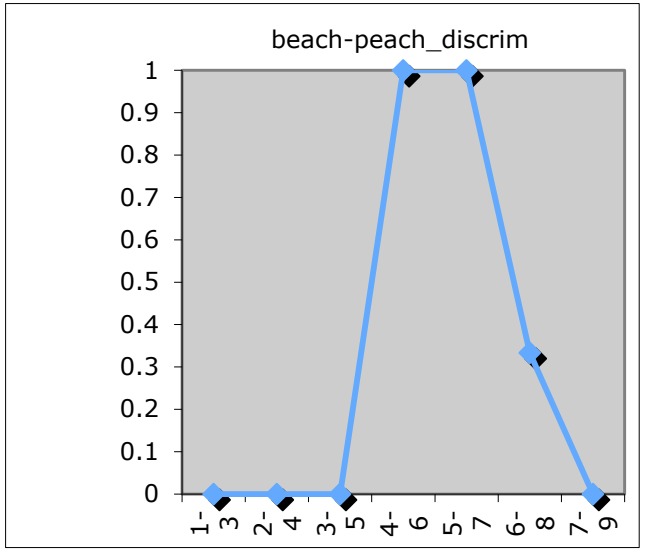


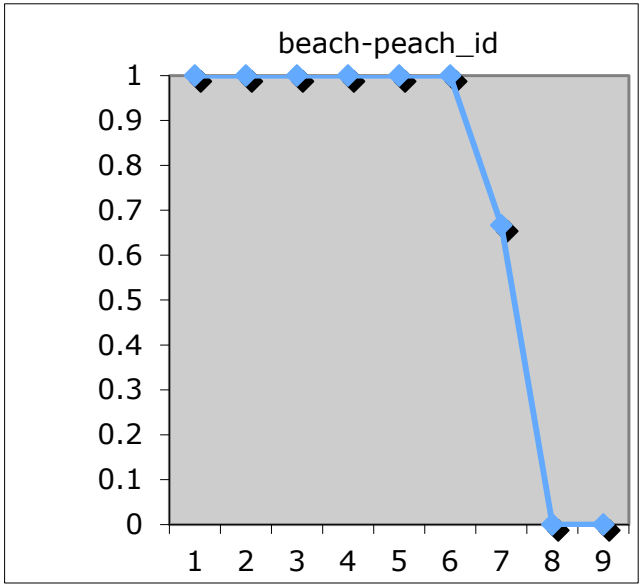


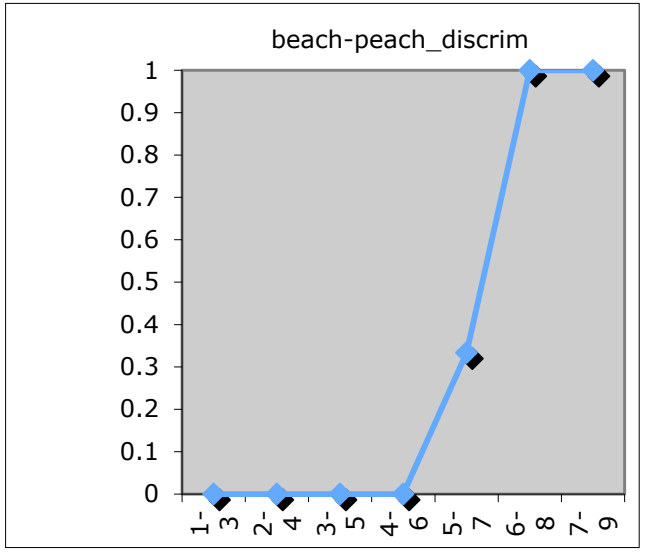










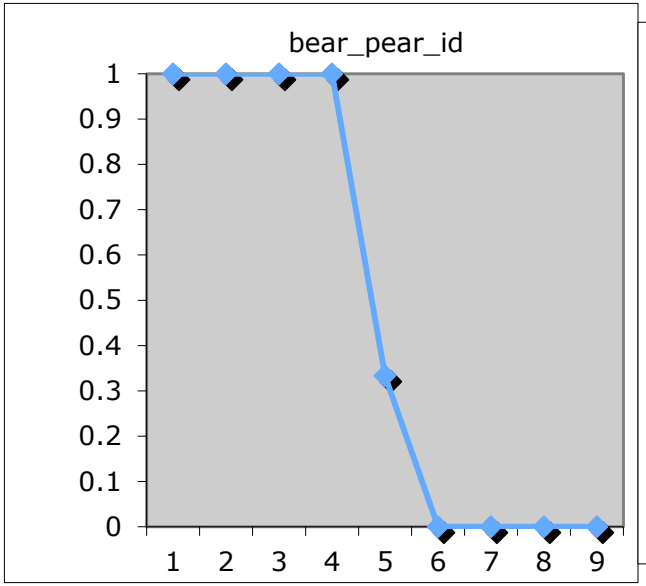


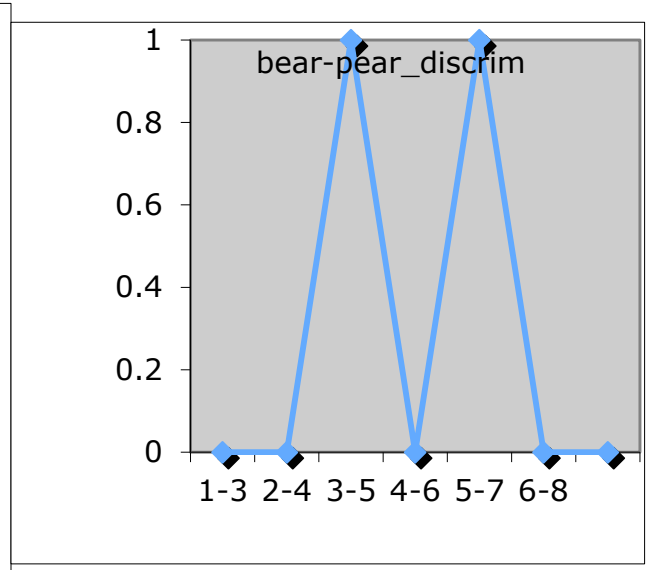
Some thoughts:

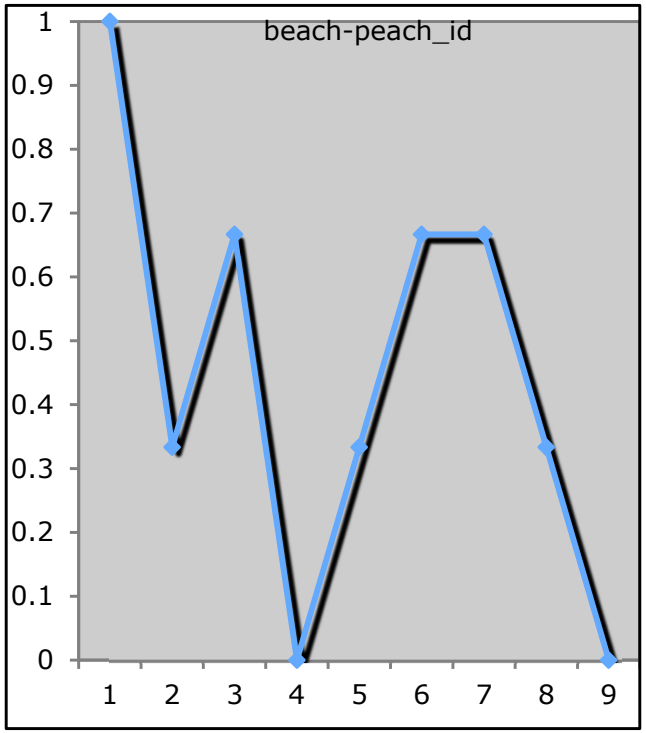
Not everyone has exactly the same category boundary.

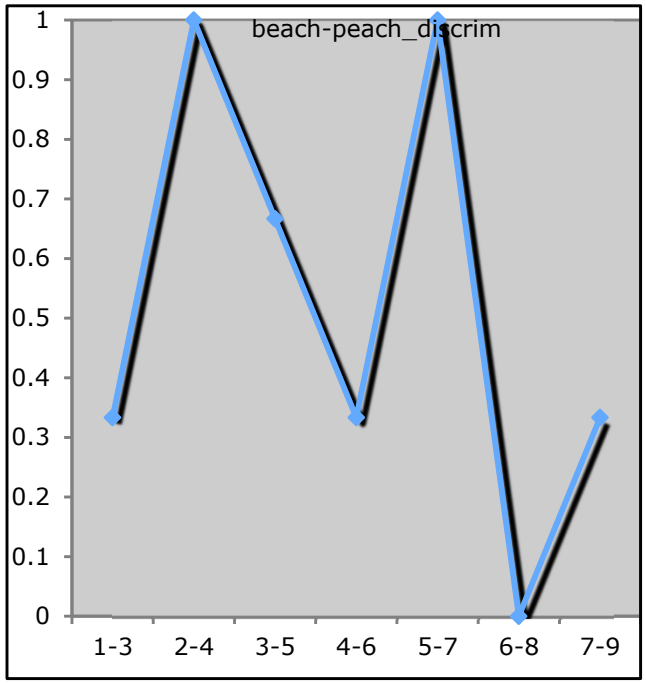
But in general, discrimination is best for stimuli perceived as being in different categories.

BTW, to make the lab do-able in a short amount of time, I only included *different* trials in the Discrim experiment! The "correct" answer on every trial would be "different," if you were able to discriminate them!









The one iron-clad law of experimental research:

Stuff happens.

Can you think of some ways this experiment was maybe not the best?

How would you do it differently?

How would categorical perception help?

OR:

What is the advantage of being *worse* at hearing some differences between sounds?

For next session

Read:

Ganong (1980)

Download here: http://search.proquest.com.libproxy.usc.edu/docview/614330814
(you'll be prompted for USC credentials)

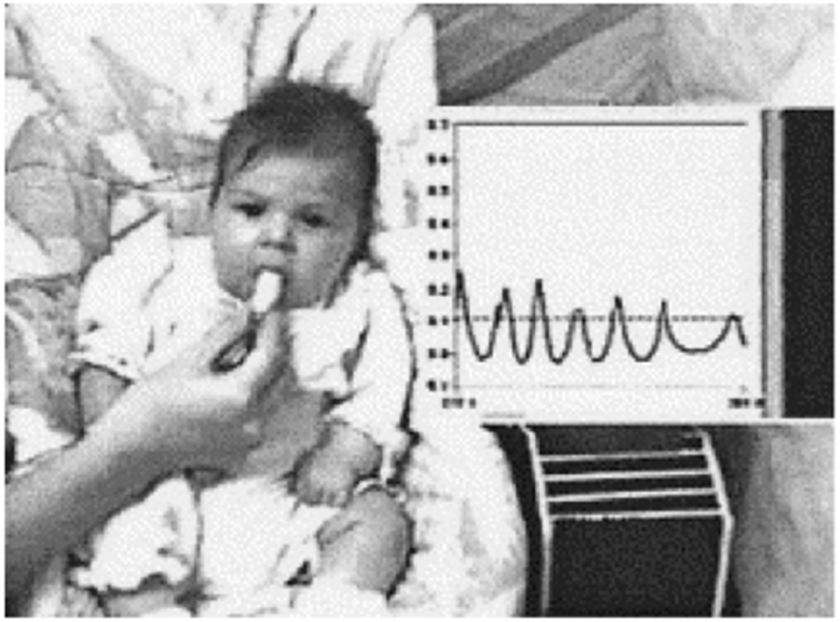
And McClelland et al. (2006)

http://psych.stanford.edu/~jlm/papers/McCMirmanHolt06_TiCS.pdf
(Prof. McClelland posts all of his papers for free on his website!)

Categorical perception in infants



Infant in the "conditioned head turn procedure from Janet Werker's website:
http://infantstudies.psych.ubc.ca/



Actually, we'll be focusing on "high amplitude sucking" results. (image from an unrelated paper, Rochat & Striano, 2001, IBD)

Categorical perception in infants

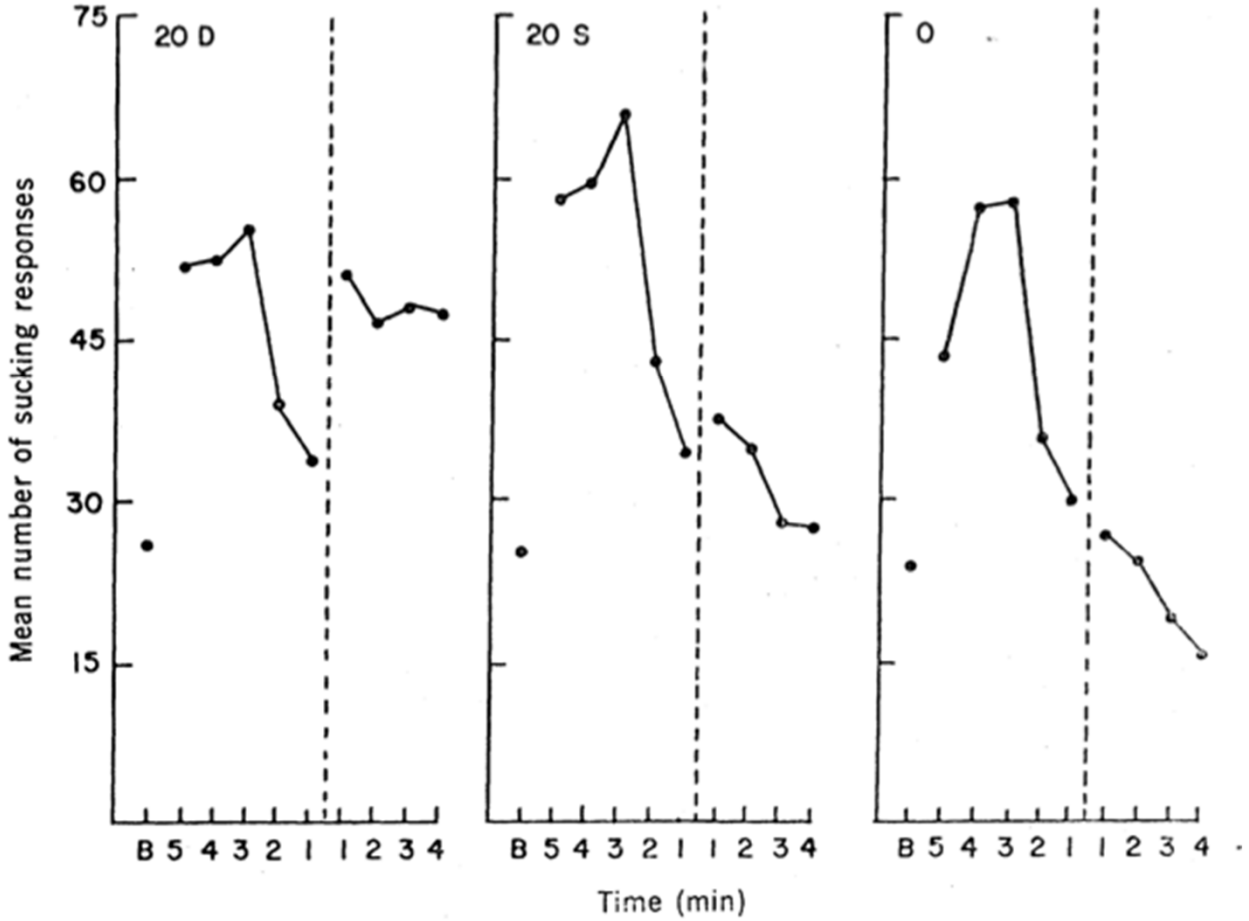
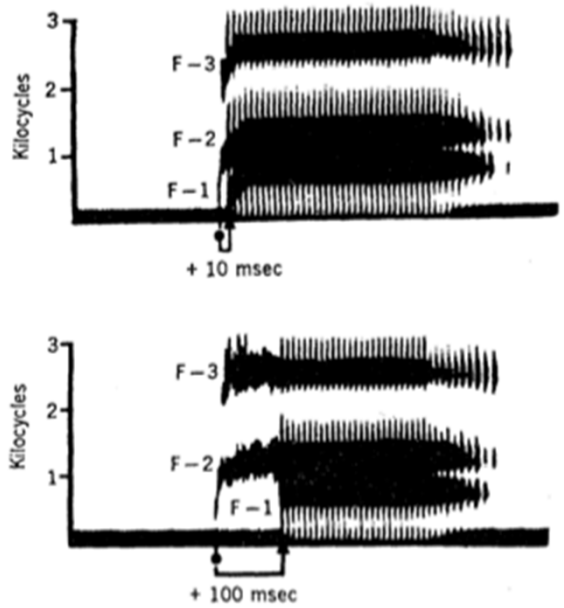


Fig. 2. Mean number of sucking responses for the 4-month-old infants, as a function of time and experimental condition. The dashed line indicates the occurrence of the stimulus shift, or in the case of the control group the time at which the shift would have occurred. The letter *B* stands for the baseline rate. Time is measured with reference to the moment of stimulus shift and indicates the 5 minutes prior to and the 4 minutes after shift.

304



In the main experiment, there were three variations in VOT differences at each of two age levels. In the first condition, 20D, the difference in VOT between the two stimuli to be discriminated was 20 msec and the two stimuli were from different adult phonemic categories. The two stimuli used in condition 20D had VOT values of +20 and +40 msec. In the second condition, 20S, the VOT difference was again 20 msec, but now the two stimuli were from the same phonemic category. In this condition the stimuli had VOT values of −20 and 0 msec or +60 and +80 msec. The third condition, 0, was a control condition in which each sub-

Eimas et al. (1971, Science)

“Perceptual narrowing” for speech categories

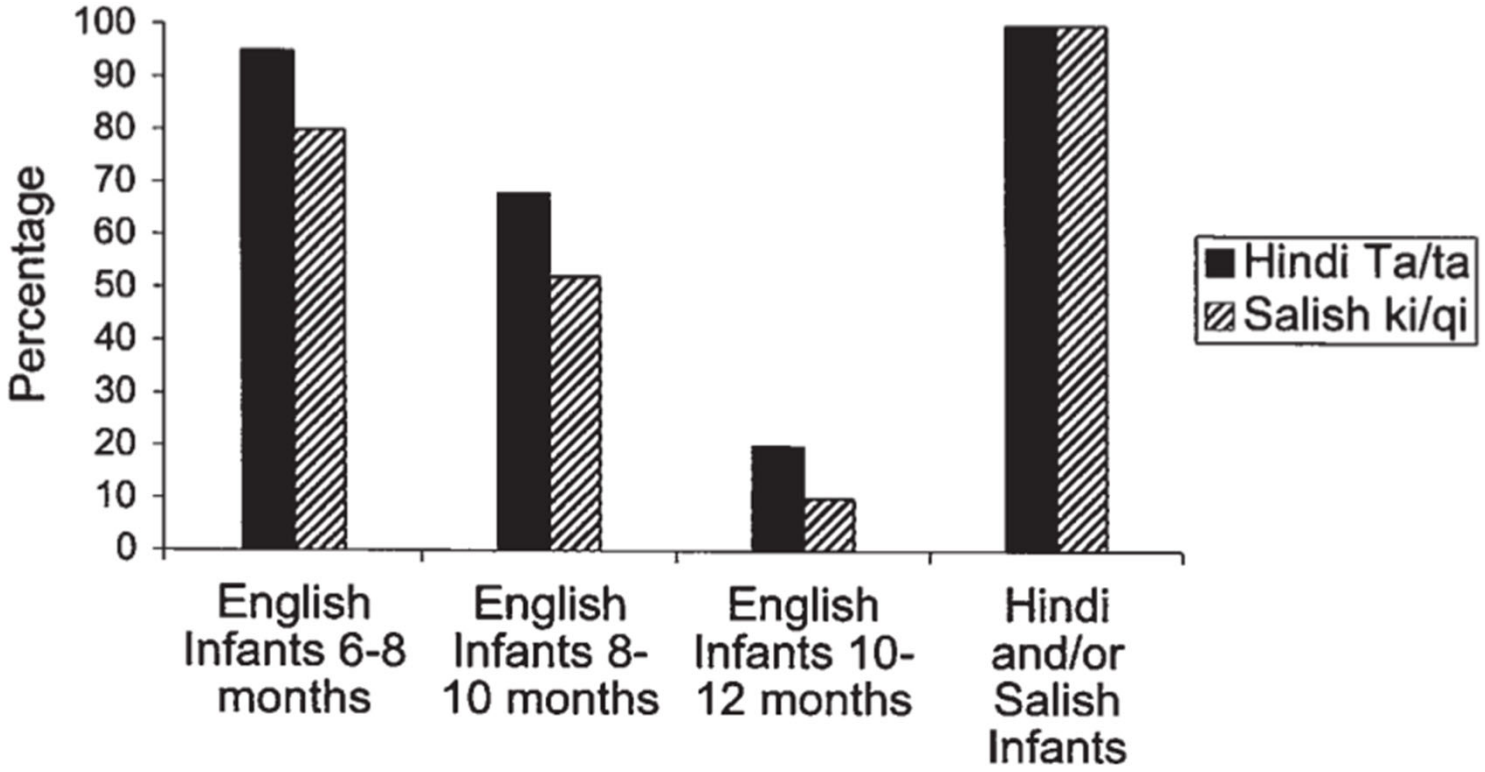


FIGURE 3 The effects of age and experience on infants' ability to discriminate two non-English contrasts. The proportion of infants at each age reaching discrimination criterion on the Hindi and Salish (non-English) contrasts. From "Cross-language speech perception: Evidence for perceptual reorganization during the first year of life," by J. F. Werker & R. C. Tees, 1984a, *Infant Behavior and Development*, 7, 49–63. Adapted with permission of the authors.

A very compelling narrative:

Speech perception depends on categorical perception abilities that are a special property of the language system.

These are present very early in development.

In fact, children start off as “citizens of the world,” able to hear all contrasts, but then rapidly tune their perceptual systems to their specific language.

Some issues

Categorical perception:

- not limited to language
- not limited to humans
- task-dependent

Infant speech perception:

- not actually “universal”

Categorical perception of non-speech sounds

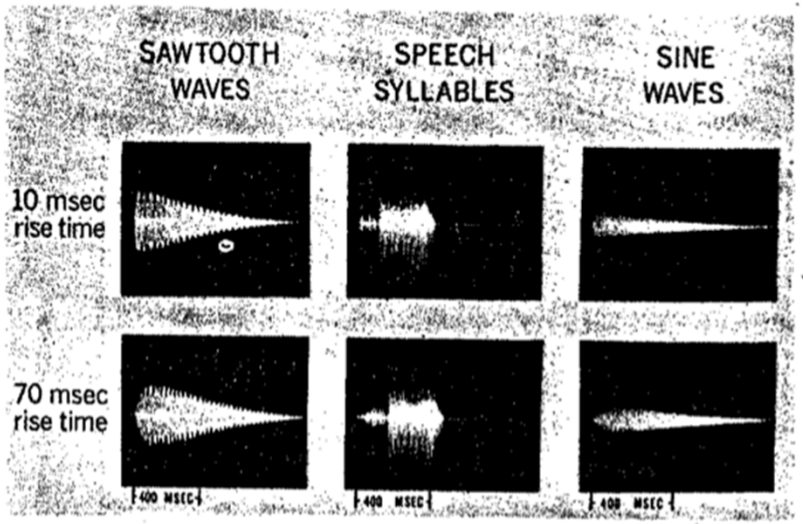
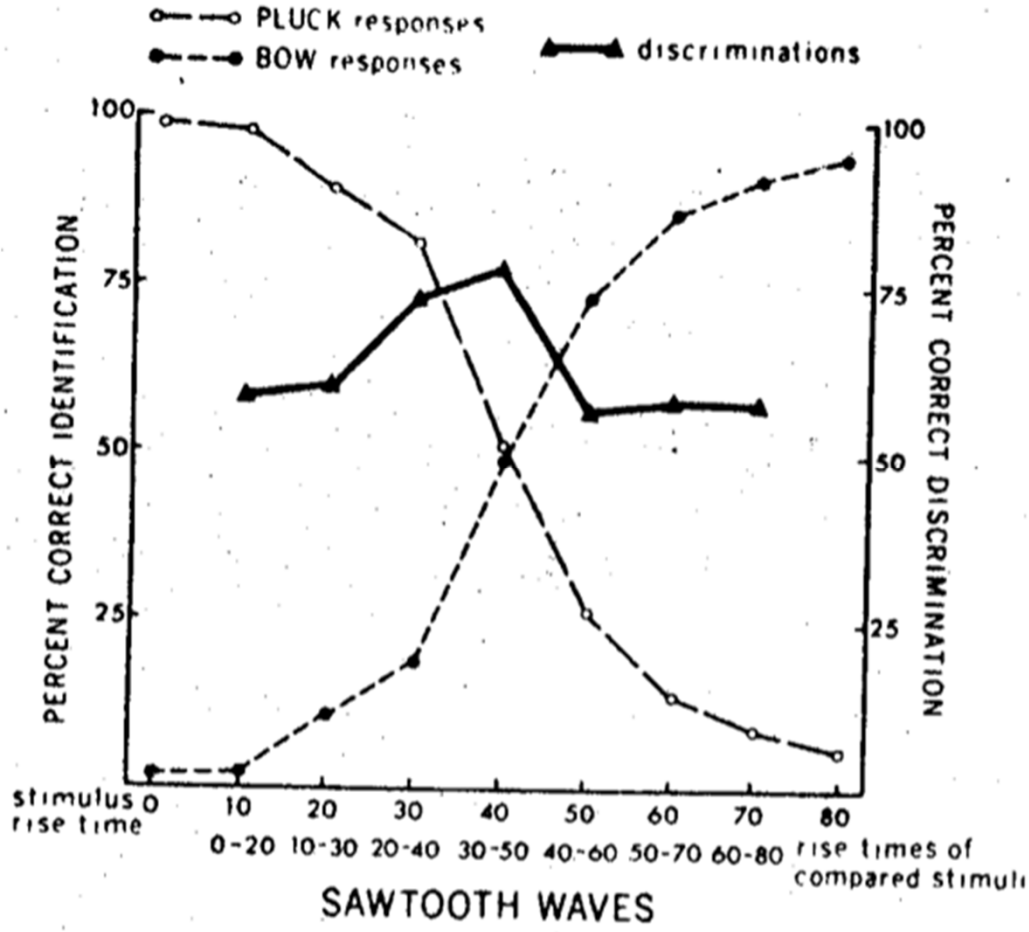


Fig. 1. Oscillograms of sample stimuli used in Experiments I and II.



Cutting & Rosner (1974, Perception & Psychophysics)

Categorical perception of non-speech sounds…by infants!

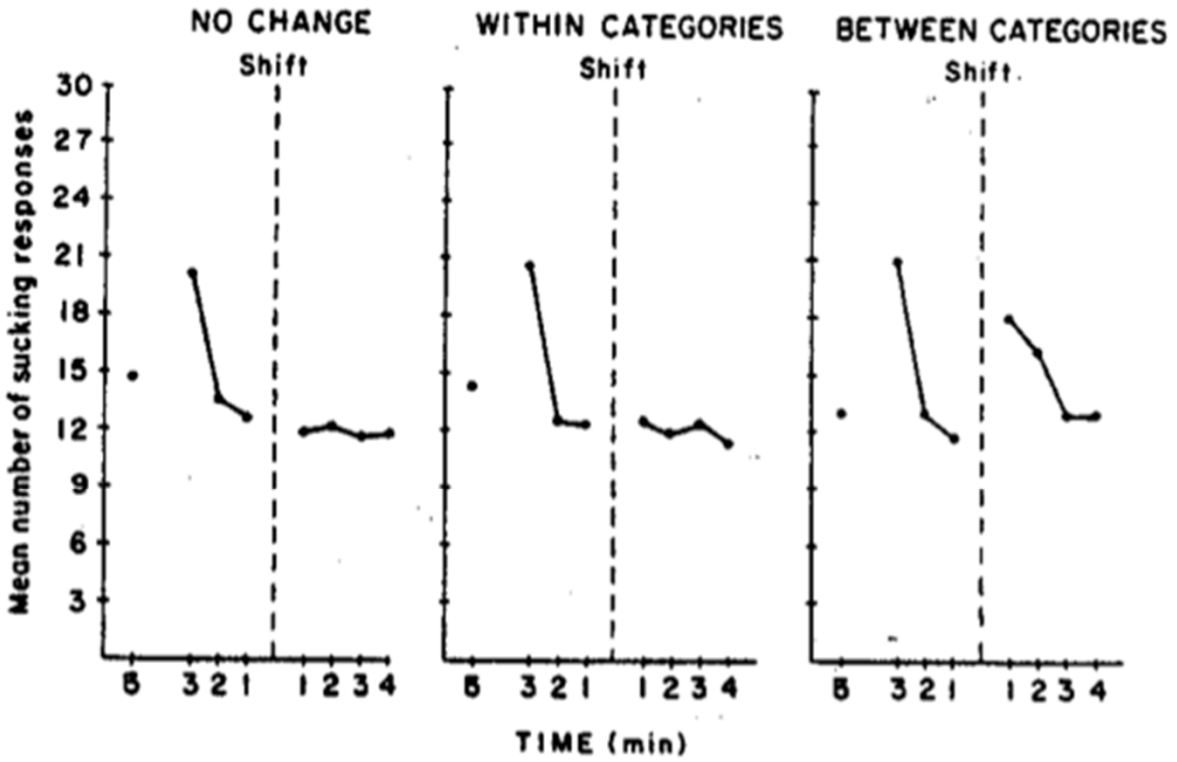


Figure 1. Mean sucking rates as a function of time and experimental session. Time is measured with reference to the moment of the stimulus shift, marked by the vertical dashed line. The baseline rate of sucking is indicated by the letter "B."

Jusczyk et al. (1977, Perception & Psychophysics)

Categorical perception of *speech* sounds…by chinchillas!



Trained using operant conditioning that either the 0-lag or 80ms-lag stimulus is a S^D for food.

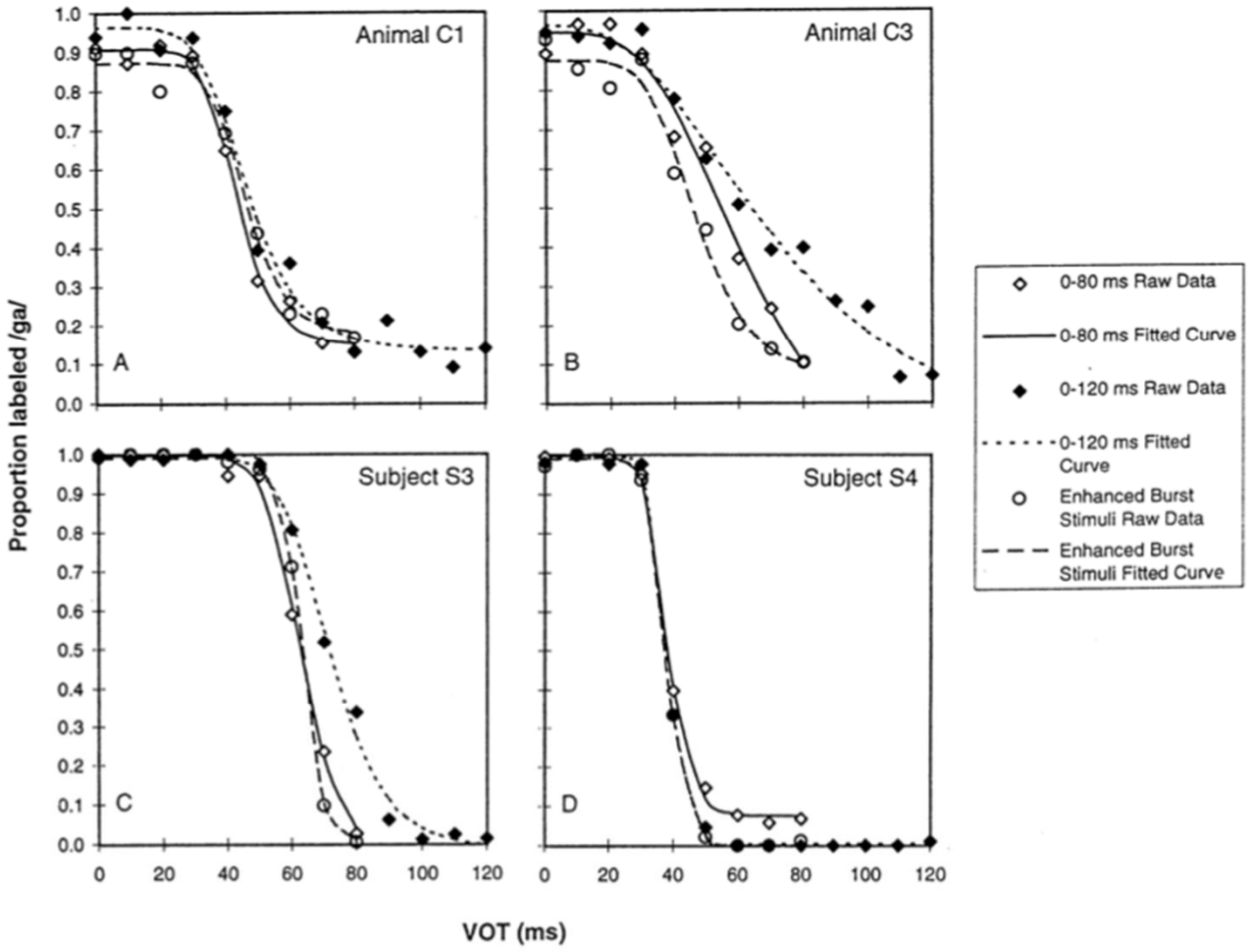


Fig. 4. Raw data and fitted curves for two chinchilla and human listeners for the /ga/–/ka/ continua. Increasing the intensity of the frication burst and aspiration noise (enhanced burst stimuli) shifted the category boundary to a shorter VOT and/or increased the slope of the curve one animal (B) and one human listener (C).

Ohlemiller et al.(1999, Behavioral Brain Research, see also Kuhl & Miller, 1975)

Some contrasts are less salient than others (for infants).

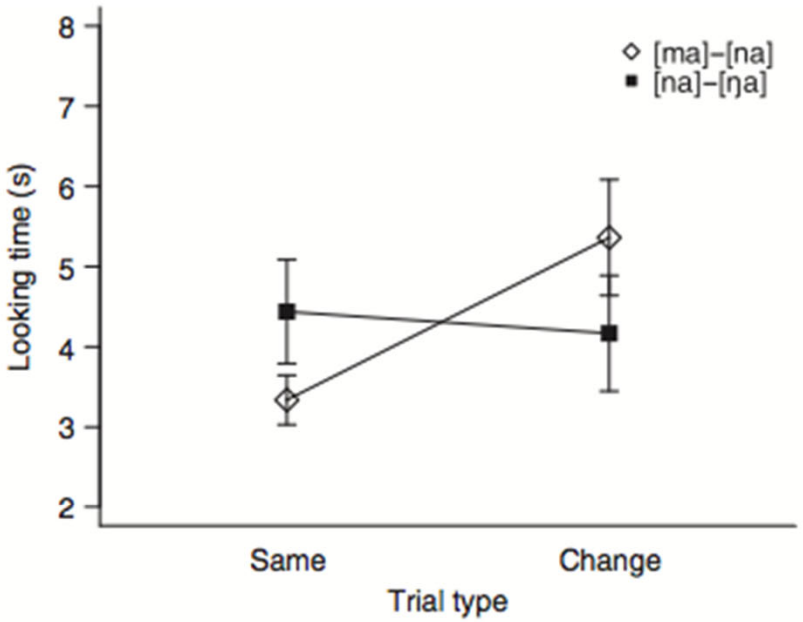


Figure 4 *English-hearing infants' (at 4–5 months) looking time to same vs. change trials for the native [ma]–[na] and non-native [na]–[ŋa] contrasts. Error bars represent standard error.*

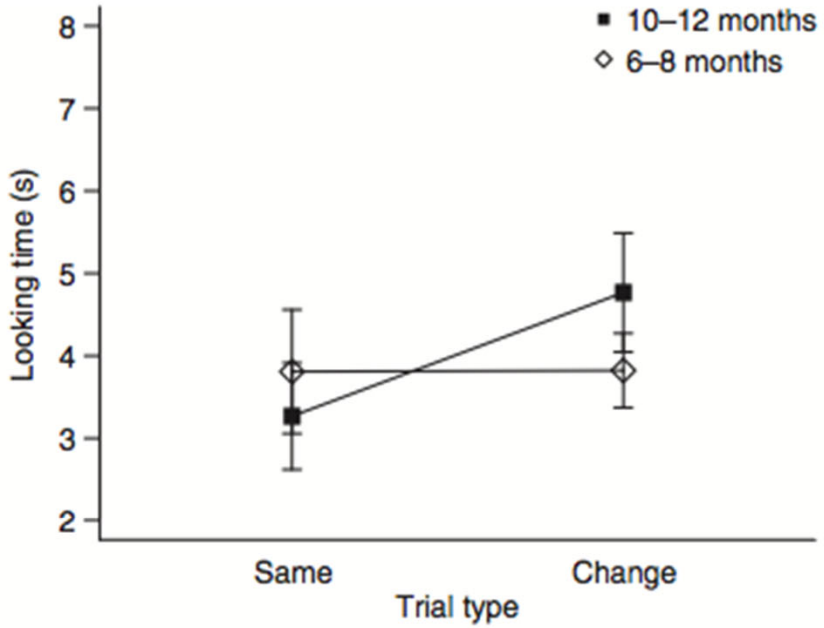


Figure 5 *Filipino-learning infants' (at 6–8 and 10–12 months) looking time to same vs. change trials for the native [na]–[ŋa]. Error bars represent standard error.*